

## Executive Summary: Sequential Training Plan Critique

### Evidence-Based Review Using 2023-2025 Literature

Review Date: January 12, 2026

Literature Base: 271 peer-reviewed papers (2023-2025)

Focus Areas: BPS Education, PIML Integration, AI-Enhanced Engineering Curricula

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#### Overall Assessment: PROMISING BUT REQUIRES MODIFICATIONS

Your Sequential Training Plan demonstrates sound pedagogical logic and aligns with emerging best practices in PIML education. However, recent literature reveals three critical gaps and one major validation concern that require attention.

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#### Four-Point Critique Summary

##### 1 Phase 1 Feasibility (Months 1-3): OPTIMISTIC

Finding: The 3-month timeline for EnergyPlus/Python automation mastery is not empirically supported by recent literature.

Evidence:

- Zakeri (2025): 86.4% project completion achieved with LLM support, but across three semesters, not 3 months
- No direct time-to-competency benchmarks found for eppy/besos in civil engineering education
- High variability in student outcomes due to knowledge gaps and time pressure

Verdict: Timeline is optimistic without stronger scaffolding

## Recommendations:

-  Require Python prerequisites OR extend Phase 1 to 4-5 months
  -  Add mandatory, graded domain checkpoints at Months 1, 2, 3
  -  Implement LLM-assisted coding with validation frameworks
  -  Conduct pilot cohort to establish empirical time-to-competency
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## 2 Domain-First Strategy: **STRONGLY SUPPORTED**

Finding: Technical evidence overwhelmingly supports domain-before-AI approach.

### Evidence:

- Jiang et al. (2024): Physics-informed ModNN achieved 156× reduction in temperature violations vs. LSTM
  - LSTM: 89°C-h violations (control failure)
  - ModNN: 0.57°C-h violations (safe operation)
- Chen et al. (2024): Domain knowledge improved robustness with small/inconsistent datasets
- Danish et al. (2025): Physics-guided models outperformed black-box ML in low-data regimes

Gap Identified: No pedagogical RCTs comparing domain-first vs. concurrent learning strategies

Verdict: Maintain domain-first core — technical evidence is very strong

## Recommendations:

-  Keep Phase 1 focused on physical domain mastery
  -  Add explicit physics-violation detection exercises in Phase 2
  -  Include case studies of ML failures due to domain ignorance
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### Curriculum Currency: GOOD (70%) — Needs Updates

Finding: Plan aligns with most 2024-2025 state-of-the-art, but missing four emerging areas.

#### Present in Plan:

- ModNN (Physics-Informed Modular Networks) — Jiang et al. (2024)
- Physics-Informed RL — Multiple 2024 papers
- Component-Based ML — Chen et al. (2024)
- LLM Code Assistance — Zakeri (2025), Maalek (2024)

#### Missing/Underrepresented:

1. Physics-Informed GNNs (Shan et al., 2025) — for urban-scale generalization
2. Neuro-Symbolic Methods (Chen et al., 2024) — Kolmogorov-Arnold Networks for equation discovery
3. Co-Simulation Frameworks (Ashayeri et al., 2024) — EnergyPlus + CONTAM integration
4. Foundation Model Safety — hallucination detection, constraint validation

Verdict: Curriculum is current but incomplete

## Recommendations:

- 🔥 Priority 1: Add Physics-Informed GNN module (Month 5)
  - 🔥 Priority 2: Include neuro-symbolic AI for interpretability (Month 6)
  - 🔥 Priority 3: Add co-simulation tooling (Month 3)
  - 🔥 Priority 4: Integrate foundation model safety protocols throughout
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## 4 Pedagogical Progression: LOGICAL BUT INCOMPLETE ⚠️

Finding: Scripting → AI Function Calling progression is sound, but critical intermediate steps are missing.

## Evidence:

- Zakeri (2025): LLMs facilitate code generation BUT require scaffolding and checkpoints
- Alphas & Tambo (2024): Designed learning environment for coding + prompting (not sequential)
- Maalek (2024): LLM support improved MATLAB performance but needed instructor oversight

## Critical Gap Identified:

Month 2: Manual Scripting → [??? 4-MONTH GAP ???] → Month 6: AI Function Calling

Verdict: Progression is pedagogically logical but lacks scaffolding

## Recommendations:

- ☒ Month 3-4: Add "AI Testing & Validation" module
  - Unit tests for AI-generated code
  - Physics-constraint validation exercises
  - Automated CI/CD for student projects
- ☒ Month 4-5: Add "Prompt Engineering for Engineers" curriculum
  - Domain-specific prompt patterns
  - Few-shot learning with BPS examples
  - Hallucination detection techniques
- ☒ Month 5-6: Add "Human-in-the-Loop Design Principles"
  - When to trust AI vs. manual override
  - Agentic workflow design patterns
  - Safety protocols for autonomous systems

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## **Priority Action Items**

### **Immediate (Before Pilot)**

1. ☒ Extend Phase 1 to 4-5 months OR add Python prerequisites
2. ☒ Design mandatory checkpoints at Months 1, 2, 3, 5, 6
3. ☒ Add intermediate modules: AI Testing (M3-4), Prompt Engineering (M4-5), HITL Design (M5-6)

## High Priority (Curriculum Updates)

4. ☒ Add Physics-Informed GNN module (Shan et al., 2025)
5. ☒ Include neuro-symbolic AI methods (Chen et al., 2024)
6. ☒ Integrate co-simulation frameworks (Ashayeri et al., 2024)

## Medium Priority (Validation)

7. ☒ Conduct pilot cohort with time-to-competency tracking
8. ☒ Develop physics-violation detection rubrics
9. ☒ Create LLM safety protocols and validation frameworks

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## Key References (Top 5)

1. Jiang, Z., & Dong, B. (2024). Modularized Neural Network Incorporating Physical Priors for Smart Building Control. *arXiv:2412.02943*. [156× reduction in violations]
2. Zakeri, B. (2025). Generative AI for Project-Based Assessment in Energy and Sustainability Education. *Preprint*. [86.4% completion with LLM support]
3. Chen, X., Singh, M. M., & Geyer, P. (2024). Utilizing Domain Knowledge: Robust Machine Learning for Building Energy Performance. *SSRN 4726883*. [Small-data robustness]
4. Shan, R., et al. (2025). Physics-Informed and Explainable Graph Neural Networks for Generalizable Urban Building Energy Modeling. *Applied Sciences*, 15(16), 8854. [State-of-the-art GNN]

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5. Alphas, R. A., & Tambo, T. (2024). Generative AI Learning Environment for Non-CS Engineering Students. *Proc. Int.*

✓ **Final Recommendation** [Coding vs. prompting pedagogy]

The Sequential Training Plan is pedagogically sound and technically current (70%), but requires:

1. Timeline extension or stronger prerequisites (Phase 1)
2. Intermediate scaffolding modules (Months 3-6)
3. Integration of four emerging technologies (GNNs, neuro-symbolic AI, co-simulation, foundation model safety)
4. Empirical validation through pilot cohort

With these modifications, the plan will be well-aligned with 2026 state-of-the-art PIML education.

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Full Detailed Critique: See `comprehensive_peer_review_critique.md`

Visual Summary: See `critique_summary_visualization.png`

Literature Base: 271 papers available in `combined_comprehensive_search_results.papertable`